2.6.10 From Pseudo-code to implementation

We have seen how a d-way heap works in a language-agnostic way. Pseudo-code provides a good way to outline and explain a data structure, without worrying about the implementation details, and hence focusing on its behavior.

At the same time, however, pseudo-code is of little practical use: to be able to move from theory to practice, we need to choose a programming language and implement our d-way heap. Independently of what platform we choose, language-specific concerns will arise and different problematics need to be taken into consideration.

We'll provide implementations of the algorithms in this book, in the effort to give readers a way to experiment and gets their hands dirty with these data structures.

The full code, including tests, can be found in our book's repo on GitHub.

2.7 Use case: find the k largest elements

In this section we are going to describe how we can use a priority queue to keep track of the k largest elements of a set.

If we have the full set of n elements in advance, we have a few alternatives that doesn't need any auxiliary data